# Modulus-Cap-Forced Growing-Ladder Signed Cancellation

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**Abstract**

We prove an actual moving-order theorem on the triangular lower-even window of RH-350. Let $m_{k,j} = k - j$, $2 \le j \le J_k$, where $J_k \to \infty$ and $J_k = o(k)$. The direct physical coefficient has the two source-locked representations

$$p_{k,j} = \tau_{\sigma,2m_{k,j}} - a_{2m_{k,j}} = Y_{k,j} + \mathcal{P}_{k,j} - S_{k,j}.$$

The modulus-complete spectral tail gives $|\tau_{\sigma,n}| \le \sigma^{-1}2^{2-n}$, while the deterministic target envelope gives $|a_n| < 48q_*^n$ with $q_* = (r_H\lambda)^{-1}$. For $H_m = mR^{-2m}$, $R = 7/5$, and $x = (R/(r_H\sqrt{\lambda}))^2$, these separate bounds become jointly subunit on the local natural scale:

$$\limsup_{k\to\infty} \left( \sup_{2\le j\le J_k} \frac{|p_{k,j}|}{2H_{m_{k,j}}x^{m_{k,j}}} \right)^{1/k} \le \max\left\{ \frac{r_H^2\lambda^3}{4}, \frac{1}{\lambda} \right\} < 1.$$

The noisy rate is strictly below 1419857/1600000. Hence the actual normalized selected direct budget decays exponentially. Combining this with RH-350 forces the actual signed remainder to track $S - \mathcal{P}$ uniformly at the natural scale and gives

$$\mathcal{Y}_k^{\mathrm{act}} = \frac{F_{J_k-2}(a_k)}{C_M} + o(1), \qquad \liminf \mathcal{Y}_k^{\mathrm{act}} \ge \frac{1}{C_M}\left( \frac{1}{x-1} - \frac{1}{x\lambda - 1} \right) > 0.$$

Thus RH-350's aggregate small-$Y$ hypothesis is false for the actual coefficients. This is normalized selected-ladder cancellation only. The unnormalized noisy modulus majorant has root $\lambda^2(qR)^2 > 9604/7225 > 1$, so it yields no unnormalized prefix closure. The full off-alias aggregate, RH-288, Gates A–E, and every Riemann-hypothesis claim remain open.

## 1 Source-locked actual coefficient

Fix the physical constants

$$r_H = \frac{17}{20}, \qquad q = \frac{1}{2}, \qquad R = \frac{7}{5}, \qquad \frac{28}{17} < \lambda < \frac{17}{10}. \tag{1}$$

The lower inequality is the strict deterministic boundary certificate of RH-262, and the upper inequality follows from the algebraic multiplier certificate in RH-336 [1, 5]. Put

$$\beta = \frac{1}{r_H\sqrt{\lambda}}, \qquad q_* = \frac{1}{r_H\lambda}, \qquad H_m = mR^{-2m}, \qquad x = (\beta R)^2. \tag{2}$$

Then

$$x\lambda = \left(\frac{R}{r_H}\right)^2 = \left(\frac{28}{17}\right)^2 > 2, \qquad x > 1, \qquad q_*R = \frac{28}{17\lambda} < 1. \tag{3}$$

Let $\sigma = \sigma_k \downarrow 0$ follow the physical natural clock and define its bounded phase by

$$\eta_k = k - \frac{\log(1/\sigma_k)}{2\log\lambda}, \qquad \sup_k |\eta_k| < \infty. \tag{4}$$

Thus, exactly,

$$\sigma_k^{-1} = \lambda^{2(k-\eta_k)}. \tag{5}$$

Take any integer-valued depth satisfying

$$J_k \longrightarrow \infty, \qquad J_k = o(k), \qquad J_k \ge 3, \tag{6}$$

and set

$$m_{k,j} = k - j, \qquad n_{k,j} = 2m_{k,j}, \qquad 2 \le j \le J_k. \tag{7}$$

Since $k - J_k \to \infty$, the whole window eventually lies in the physical RH-348 ladder.

RH-282 constructs the modulus-complete normal spectral complement $C_\sigma$ and proves

$$\tau_{\sigma,n} = \operatorname{Tr} C_\sigma^n, \qquad |\tau_{\sigma,n}| \le \sigma^{-1} q^{n-2} \quad (n \ge 2). \tag{8}$$

RH-267 proves the all-order deterministic numerator bound

$$|a_n| < 48 q_*^n \quad (n \ge 2). \tag{9}$$

These are different source estimates, but RH-288 and RH-340 identify their difference as one actual direct coefficient:

$$p_{\sigma,k,n} = \tau_{\sigma,n} - a_n. \tag{10}$$

Equations (8)–(10) are respectively the modulus-complement, deterministic-target, and direct-coefficient data types; no head/counterloop substitution is made [2, 7–9].

On the lower-even ladder, RH-348 supplies the second exact representation

$$\boxed{p_{k,j} := p_{\sigma,k,2m_{k,j}} = Y_{k,j} + \mathcal{P}_{k,j} - S_{k,j},} \tag{11}$$

where

$$Y_{k,j} = \mathcal{T}^{\mathrm{rest}}_{k,m_{k,j}} - d_{\sigma,k,2m_{k,j}}. \tag{12}$$

This $Y$ is an actual signed physical remainder, not a free completion variable [4, 6].

Finally, RH-350 proves, uniformly on (6),

$$\epsilon_k^S := \sup_j \left| \frac{C_M S_{k,j}}{2H_{m_{k,j}} x^{m_{k,j}}} - 1 \right| \longrightarrow 0, \tag{13}$$

$$\epsilon_k^P := \sup_j \left| \frac{C_M \mathcal{P}_{k,j}}{2H_{m_{k,j}} x^{m_{k,j}}} - a_k \lambda^{2-j} \right| \longrightarrow 0, \tag{14}$$

where $C_M > 0$ is fixed and $a_k = C_* C_M \lambda^{\eta_k - 2} > 0$ [3]. The new input below is not another estimate of $S$ or $\mathcal{P}$; it is the direct bound on the actual $p$ supplied by (8) and (9).

## 2 Exponential local natural-scale theorem

Define

$$U_k = \sup_{2 \le j \le J_k} \frac{|p_{k,j}|}{2H_{m_{k,j}} x^{m_{k,j}}}, \tag{15}$$

and the two fixed root rates

$$\rho_N = \frac{r_H^2 \lambda^3}{4}, \qquad \rho_T = \frac{1}{\lambda}. \tag{16}$$

**Theorem 2.1** (actual uniform natural-scale cancellation)**.** *For every clock and growing window satisfying* (4)–(6),

$$\boxed{\limsup_{k\to\infty} U_k^{1/k} \le \max\{\rho_N, \rho_T\} < 1.} \tag{17}$$

*More precisely,*

$$\rho_N < \frac{1419857}{1600000} < 1, \qquad \rho_T < \frac{17}{28} < 1. \tag{18}$$

*Proof.* Write $m = m_{k,j}$. Since $2H_m x^m = 2m\beta^{2m}$, (8) gives

$$\begin{aligned}\frac{|\tau_{\sigma,2m}|}{2H_m x^m} &\le \frac{\sigma^{-1}q^{2m-2}}{2m\beta^{2m}} \\ &= \frac{2}{m}\sigma^{-1}(q^2 r_H^2 \lambda)^m. \end{aligned} \tag{19}$$

Put

$$b_N = q^2 r_H^2 \lambda. \tag{20}$$

The upper bound on $\lambda$ gives $b_N < 4913/16000 < 1$. Using (5) and $m = k - j$, the right side of (19) becomes

$$\frac{2\lambda^{-2\eta_k}}{m}\rho_N^k b_N^{-j}. \tag{21}$$

Here $m \ge k - J_k$, the phase factor is bounded, and $b_N^{-J_k} = \exp(o(k))$. Therefore

$$\limsup_{k\to\infty}\left(\sup_j \frac{|\tau_{\sigma,2m_{k,j}}|}{2H_{m_{k,j}}x^{m_{k,j}}}\right)^{1/k} \le \rho_N. \tag{22}$$

For the target, (9) and $q_*^2/\beta^2 = 1/\lambda$ give

$$\frac{|a_{2m}|}{2H_m x^m} < \frac{24}{m}\left(\frac{1}{\lambda}\right)^m = \frac{24}{m}\lambda^{-k}\lambda^j. \tag{23}$$

Since $j \le J_k = o(k)$,

$$\limsup_{k\to\infty}\left(\sup_j \frac{|a_{2m_{k,j}}|}{2H_{m_{k,j}}x^{m_{k,j}}}\right)^{1/k} \le \rho_T. \tag{24}$$

The triangle inequality in (10) and the standard root bound for a sum of nonnegative sequences prove (17).

Finally, $r_H = 17/20$ and $\lambda < 17/10$ imply

$$\rho_N < \frac{(17/20)^2(17/10)^3}{4} = \frac{1419857}{1600000} < 1.$$

The lower bound $\lambda > 28/17$ gives $\rho_T < 17/28 < 1$. □

The theorem is genuinely uniform on a moving window. The factors generated by $J_k$ are subexponential because $J_k = o(k)$; no finite collection of fixed-$j$ limits is promoted to a growing-order claim.

Define the actual normalized selected direct budget

$$\mathcal{L}_k^{\mathrm{act}} = \frac{1}{x^{k-2}}\sum_{j=2}^{J_k}\frac{|p_{k,j}|}{2H_{m_{k,j}}}. \tag{25}$$

**Corollary 2.2** (actual normalized selected-budget closure)**.** *One has*

$$\boxed{\limsup_{k\to\infty}(\mathcal{L}_k^{\mathrm{act}})^{1/k} \le \max\{\rho_N,\rho_T\} < 1,} \tag{26}$$

*and hence $\mathcal{L}_k^{\mathrm{act}} \to 0$ exponentially.*

*Proof.* Since $m_{k,j} = k - j$,

$$\begin{aligned}\mathcal{L}_k^{\mathrm{act}} &= \sum_{j=2}^{J_k} x^{2-j}\frac{|p_{k,j}|}{2H_{m_{k,j}}x^{m_{k,j}}} \\ &\le U_k \sum_{j=2}^{\infty} x^{2-j} = \frac{U_k}{1-x^{-1}}.\end{aligned}$$

The fixed geometric factor has $k$th root tending to one, so theorem 2.1 proves the result. □

## 3 Forced actual signed completion

The exact identity (11) can now be solved for the actual remainder without changing data type:

$$Y_{k,j} = S_{k,j} - \mathcal{P}_{k,j} + p_{k,j}. \tag{27}$$

**Theorem 3.1** (uniform actual close-completion law)**.** *On every window* (6)*,*

$$\boxed{\sup_{2\le j\le J_k}\left|\frac{C_M Y_{k,j}}{2H_{m_{k,j}}x^{m_{k,j}}} - (1-a_k\lambda^{2-j})\right| \longrightarrow 0.} \tag{28}$$

*Thus the actual $Y$ array asymptotically tracks the RH-351 close completion $S-\mathcal{P}$ at the local natural scale.*

*Proof.* Substitute (27) and apply the triangle inequality. The left side of (28) is at most

$$\epsilon_k^S + \epsilon_k^P + C_M U_k.$$

The first two terms vanish by (13)–(14); the last vanishes exponentially by theorem 2.1. □

For $N \ge 0$ recall the RH-350 weighted functional

$$F_N(a) = \sum_{r=0}^{N} x^{-r}|a\lambda^{-r} - 1|. \tag{29}$$

Its exact minimum is

$$A_N := \inf_{a>0} F_N(a) = \frac{1-x^{-N}}{x-1} - \frac{1-(x\lambda)^{-N}}{x\lambda - 1}, \tag{30}$$

$$A_N \nearrow A_\infty = \frac{1}{x-1} - \frac{1}{x\lambda-1} > 0. \tag{31}$$

Define the actual normalized selected signed-remainder budget

$$\mathcal{Y}_k^{\mathrm{act}} = \frac{1}{x^{k-2}}\sum_{j=2}^{J_k}\frac{|Y_{k,j}|}{2H_{m_{k,j}}}. \tag{32}$$

**Corollary 3.2** (actual aggregate law and failure of small $Y$)**.** *With $N_k = J_k - 2$,*

$$\boxed{\mathcal{Y}_k^{\mathrm{act}} = \frac{F_{N_k}(a_k)}{C_M} + o(1),} \tag{33}$$

*and*

$$\boxed{\liminf_{k\to\infty}\mathcal{Y}_k^{\mathrm{act}} \ge \frac{A_\infty}{C_M} = \frac{1}{C_M}\left(\frac{1}{x-1} - \frac{1}{x\lambda-1}\right) > 0.} \tag{34}$$

*Consequently the actual aggregate small-$Y$ hypothesis used conditionally in RH-350 is false.*

*Proof.* Let

$$z_{k,j} = \frac{C_M Y_{k,j}}{2H_{m_{k,j}}x^{m_{k,j}}}.$$

By theorem 3.1, uniformly in $j$, $z_{k,j} = 1 - a_k\lambda^{2-j} + o(1)$. Since $||z|-|w|| \le |z-w|$ and $\sum_{j=2}^{\infty} x^{2-j} < \infty$,

$$\begin{aligned}\mathcal{Y}_k^{\mathrm{act}} &= \frac{1}{C_M}\sum_{j=2}^{J_k} x^{2-j}|z_{k,j}| \\ &= \frac{1}{C_M}\sum_{j=2}^{J_k} x^{2-j}|1-a_k\lambda^{2-j}| + o(1) \\ &= \frac{F_{J_k-2}(a_k)}{C_M} + o(1).\end{aligned}$$

Because $a_k > 0$, (30) gives $F_{N_k}(a_k) \ge A_{N_k}$. Now $N_k \to \infty$, so (31) proves (34). □

This conclusion is about the actual $Y$ in (12), not a formal coefficient ledger. It nevertheless remains local to the selected lower-even window and to the normalization in (32).

## 4 Why the unnormalized prefix remains open

Remove the factor $x^m$ from the denominator in (19). The RH-282 modulus estimate then gives the explicit separate majorant

$$\frac{|\tau_{\sigma,2m}|}{2H_m} \le \widehat{N}_{k,j} := \frac{2}{m}\sigma^{-1}(qR)^{2m}. \tag{35}$$

On the same window, bounded phase and $J_k = o(k)$ give

$$\lim_{k\to\infty}\left(\sup_{2\le j\le J_k} \widehat{N}_{k,j}\right)^{1/k} = \lambda^2(qR)^2. \tag{36}$$

This rate is strictly superunit:

$$\lambda^2(qR)^2 > \left(\frac{28}{17}\right)^2\left(\frac{7}{10}\right)^2 = \frac{9604}{7225} > 1. \tag{37}$$

By contrast, the separate target majorant has unnormalized root $(q_*R)^2 < 1$. Equivalently,

$$x\rho_N = \lambda^2(qR)^2 > 1. \tag{38}$$

Equations (36)–(38) are a method boundary, not a lower bound for the actual coefficient. The actual difference $p = \tau - a$ may contain additional signed or complex cancellation, but the two separate absolute source bounds do not reveal it. Therefore theorem 2.1 and corollary 2.2 cannot be multiplied by the removed $x^k$ scale to conclude that the unnormalized selected prefix tends to zero. Such a conclusion requires a stronger theorem directly on $\tau - a$.

## 5 Executable protocol and claim boundary

The executable artifact performs exact rational checks of (3), (18), and (37). At the in-range fixture $\lambda = 5/3$ and $\eta_k = 0$, it reproduces the finite coordinate majorants (19) and (23) on windows $J = \lfloor\sqrt{k}\rfloor$. It also checks the finite algebra $Y = S - \mathcal{P} + p$. These rows are formula-reproduction checks only: they are not observations of $\tau$, $a$, $p$, or $Y$, are not interval certificates for the physical multiplier, and are not asymptotic evidence.

RH-352 proves actual normalized/local natural-scale signed cancellation on one growing lower-even ladder. It does not prove that the unnormalized selected prefix vanishes; control the critical, first-lower, odd, or upper-alias orders; decide the full $E_{\mathrm{off}}$ aggregate; or prove the RH-241 moving noisy all-order envelope and coefficient bridge. It does not activate the RH-288 determinant-gluing criterion. Gates A–E remain false/open. This paper constructs no Hilbert–Polya operator, identifies no Riemann zero, proves no von Mangoldt prime-power trace, proves no completed-zeta divisor equality, and does not prove the Riemann Hypothesis.

## References


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